COURSE & ATTENDANCE MANAGEMENT SYSTEM

Project Scope & Development Proposal

1. Project Overview

The proposed system is a web-based **Course & Attendance Management System** designed to simplify the management of teachers, students, courses, course enrollment, and attendance.

The platform will provide separate interfaces and permissions for the **Admin, Teachers, and Students**, allowing each user type to perform the activities relevant to their role.

The initial system will support **six courses**, with the ability for teachers to create additional courses as the system grows.

2. User Roles

The system will have three user roles:

A. Admin

The Admin oversees the entire platform and manages users and system activities.

B. Teacher

Teachers manage the courses they teach and record attendance for students enrolled in those courses.

C. Student

Students can view available courses, enroll in courses, and view their attendance records. Payments made for enrollment will be handled and confirmed by the institution.

3. Admin Features

The Admin will be able to:

Dashboard

- View an overview of the system

- View total number of teachers
- View total number of students
- View total number of courses
- View relevant enrollment/attendance information

Teacher Management

- Onboard/create teacher accounts
- View the list of all teachers
- View individual teacher details
- View courses assigned to/created by a teacher
- Manage teacher information

Student Management

- Onboard/create student accounts
- View the list of all students
- View individual student details
- View courses registered by each student
- Manage student information

Course Management / Monitoring

- View all available courses
- View course details
- View the teacher who created/teaches each course
- View enrolled students
- Monitor course activity

Courses created by teachers will automatically become part of the system's available course catalogue.

4. Teacher Features

Teachers will have authority over the courses they create and teach.

Teacher Dashboard

- View an overview of their courses
- View number of enrolled students
- View relevant attendance information

Course Management

Teachers can:

- Create a new course
- Provide course name/details
- Set the fixed weekly day for the course
- Set the course start and end time
- View courses they teach
- View individual course details
- View students enrolled in their courses

Student Enrollment Visibility

When students enroll in a teacher's course, the enrollment will automatically appear under that teacher's course.

For example:

Teacher creates **Data Analysis**
↓
Data Analysis appears in Available Courses
↓
Student enrolls
↓
Student automatically appears under Data Analysis
↓
Teacher can see the enrolled student

Teachers will not need to manually add students to their courses.

Attendance

Teachers will be able to:

- Create/view class sessions
- Mark student attendance
- View attendance history
- View attendance records for their courses
- View attendance percentage where applicable

5. Student Features

Students will be able to:

Student Dashboard

- View an overview of their registered courses
- View upcoming course schedules
- View attendance information

Course Catalogue

- View available courses
- View course details
- View course schedule
- View the teacher teaching the course

Course Enrollment

- Enroll in multiple courses
- View courses they have registered for
- View their course schedules

Students may register for as many available courses as they are eligible for.

Attendance

Students will be able to:

- View their attendance history
- View attendance records by course
- View attendance percentage where applicable

6. Course Rules

The system will follow these rules:

- Teachers have authority to create courses.
- A course is automatically associated with the teacher who creates it.
- Newly created courses automatically appear in the available course catalogue.
- Students can enroll in multiple courses.
- When a student enrolls in a course, the enrollment is automatically associated with that course and its teacher.
- Each course has a fixed weekly day and time.
- The initial system is designed around six courses but can support additional courses.
- Payment for courses is handled **outside the system** and is not part of the project.

7. Attendance Management

The system will maintain attendance records based on individual class sessions.

Teachers will mark whether each enrolled student is present or absent.

The system will store attendance history so that:

Teacher → can view attendance for their courses

and

Student → can view their own attendance history.

Attendance percentages can also be calculated from the stored attendance records.

8. System Requirements

The application will include:

- Secure authentication/login
- Role-based access
- Database
- API/backend
- Course management
- Student enrollment
- Teacher management
- Student management
- Attendance management
- Data validation
- Loading states
- Empty states
- Error handling
- Responsive design for desktop, tablet and mobile
- Deployment of the completed application

Additional features requested after scope approval may require a separate cost and/or timeline adjustment.

10. Project Timeline

The estimated development timeline is **4–5 weeks**.

Phase	Activities	Estimated Duration
1. Requirements & Scope Finalization	Confirm requirements, workflows, roles and acceptance criteria	2–3 days
2. UI/UX Design	User flows, wireframes, Figma designs and responsive layouts	4–5 days
3. Backend & Database	Database structure, authentication, APIs and role permissions	5–7 days
4. Frontend Development	Admin, Teacher & Student interfaces and API integration	7–10 days
5. Testing & Refinement	Functional testing, bug fixing, responsiveness and UAT	3–4 days
6. Deployment	Application/backend/database deployment and final configuration	1–2 days

Estimated Total:

Approximately 4–5 weeks from scope approval and initial payment.

The timeline may be adjusted if there are delays in receiving feedback, content, approvals, or if additional features are requested outside the agreed scope.

11. Documentation

The project will include basic documentation covering:

A. User Guide

A simple guide explaining how each user type uses the system:

- Admin login
- Onboarding teachers
- Onboarding students
- Viewing teachers/students
- Teacher course creation
- Student course enrollment
- Teacher attendance marking
- Student attendance viewing

B. Technical Documentation

Documentation covering the technical setup of the application, including:

- Project structure
- Technology stack
- Frontend setup
- Backend/API setup
- Database structure
- Environment variables/configuration
- API endpoints
- Authentication and role permissions
- Local development setup
- Deployment information

C. Database Documentation

A basic explanation of the major entities and their relationships:

Users

├── Admin

├── Teachers

└── Students

Teachers

↓

Courses

↓

Enrollments

↑

Students

↓

Courses

↓

Class Sessions

↓

Attendance Records

D. Handover Documentation

At project completion, the client receives the necessary information for operating and maintaining the application, including relevant access/setup information.

12. Deliverables

At completion, the client will receive:

- Responsive web application
- Super Admin interface
- Teacher interface
- Student interface

- Authentication system
- Course management system
- Enrollment system
- Attendance management system
- Database
- Backend/API
- Testing and bug fixes
- Deployed application

12. Development Cost

Total Development Fee: ₦300,000

Suggested payment structure:

50% — ₦150,000 upfront
Required before development/design begins.

50% — ₦150,000 upon completion
Payable after the agreed features have been implemented, tested and deployed.

Hosting/domain/infrastructure costs, if applicable, are separate from the development fee.

13. Scope Changes

The agreed scope serves as the basis for the development timeline and cost.

Any feature or functionality requested after approval of the scope that is not included in this document will be treated as an additional requirement and may result in:

- Additional development cost
- Additional development time

The revised cost/timeline will be communicated and agreed upon before the additional work begins.

14. Project Process

The project will follow this process:

Requirements & Scope
↓

Client Approval
↓
Initial Payment
↓
UI/UX Design / Figma
↓
Database & Backend
↓
Frontend Development
↓
Integration
↓
Testing & User Acceptance
↓
Deployment
↓
Project Completion